DATA ANALYTICS INTERNSHIP TASK

TOPIC:

EXCHANGE RATE IMPACT ON IMPORTS IN PAKISTAN

SUBMITTED BY:

MUHAMMAD ABDULLAH HASSAN

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OPTIMUS AUTOMATE

Title: Exchange Rate Impact on Imports in Pakistan

Introduction:

Exchange rate plays a crucial role in determining a country's international trade performance, especially imports. Fluctuations in the exchange rate affect the cost of foreign goods and services, which directly influences the volume of imports. When the domestic currency depreciates, imported goods become more expensive, leading to a potential decline in imports. Conversely, an appreciation of the currency makes imports cheaper and may increase import volume.

In a developing country like Pakistan, imports are essential for industrial inputs, machinery, fuel, and consumer goods. Therefore, understanding how exchange rate movements affect imports is vital for policymakers to manage the trade balance and foreign reserves. This study examines the relationship between the exchange rate and imports over the period **2000–2024** using multiple regression analysis.

Objectives:

- To examine the impact of the exchange rate on imports in Pakistan.
- To identify other macroeconomic variables affecting imports.
- To estimate the relationship between imports and selected variables using regression analysis.
- To provide policy recommendations based on results.

Data Description:

Variable	Description	Source
Imports	Imports of goods and services	World Bank
Rate	Official exchange rate	World Bank
GDP	Gross Domestic Product	World Bank

Period: 2000-2024

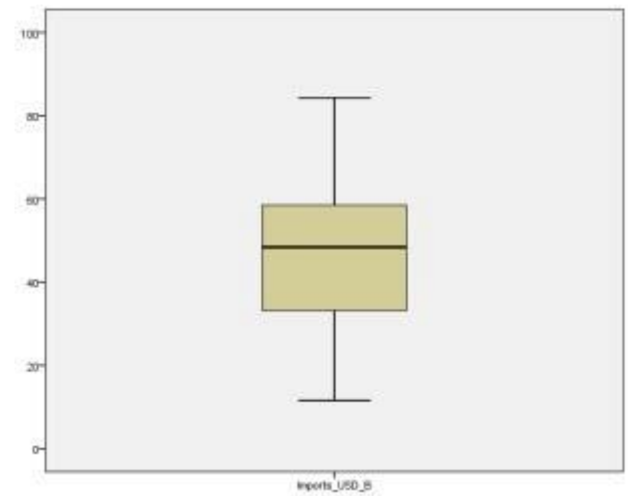
Methodology:

Tools Used:

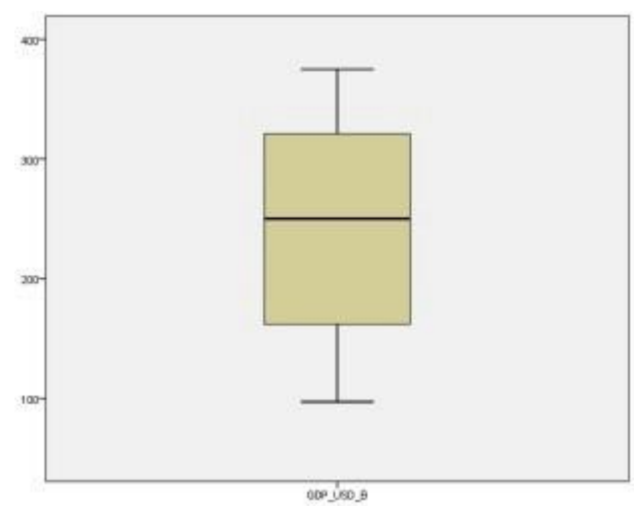
- MS Excel
- EViews
- SPSS 27

Box Plots:

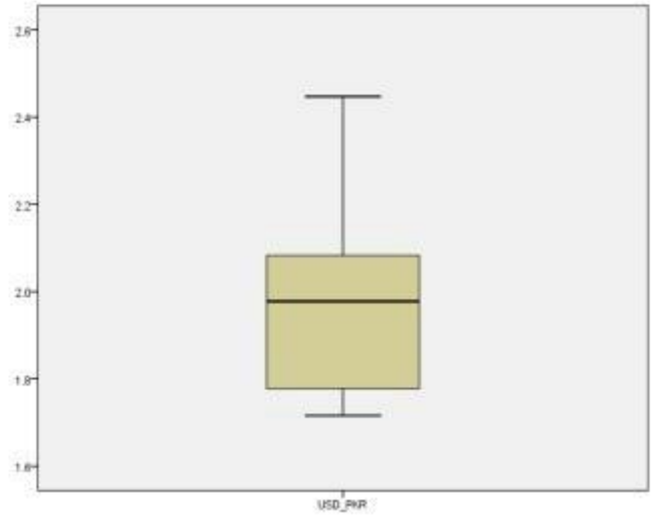
GDP Box and Whisker Plot



Imports Box and Whisker Plot



Exchange Rate Box and Whisker Plot

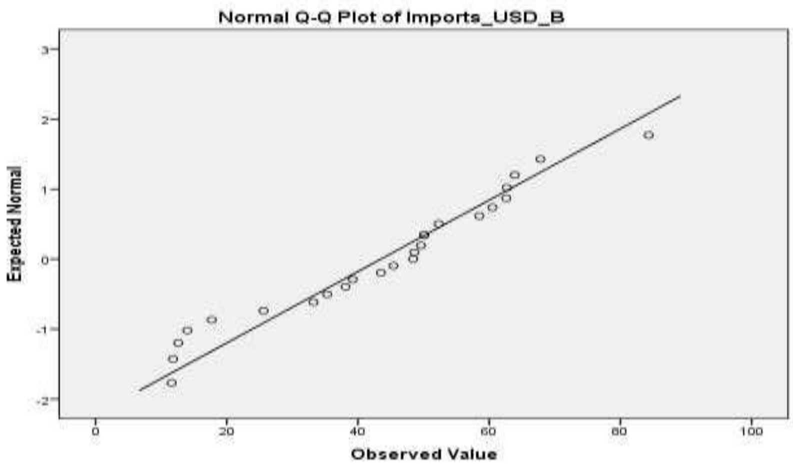


These are box-and-whisker plots, so they show median, spread (IQR), range, and possible outliers of each variable.

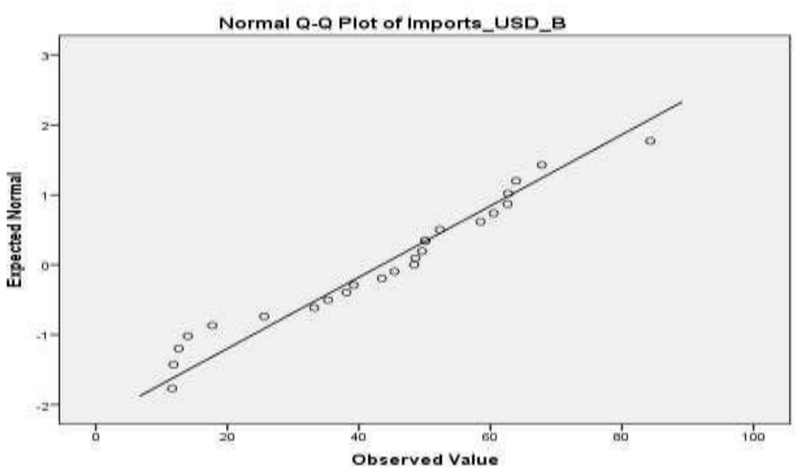
- GDP has the highest variability (the largest spread and range).
- Imports show moderate variability.
- The exchange rate is the most stable (smallest spread).

In simple words, GDP fluctuates the most, imports fluctuate moderately, and the exchange rate changes the least.

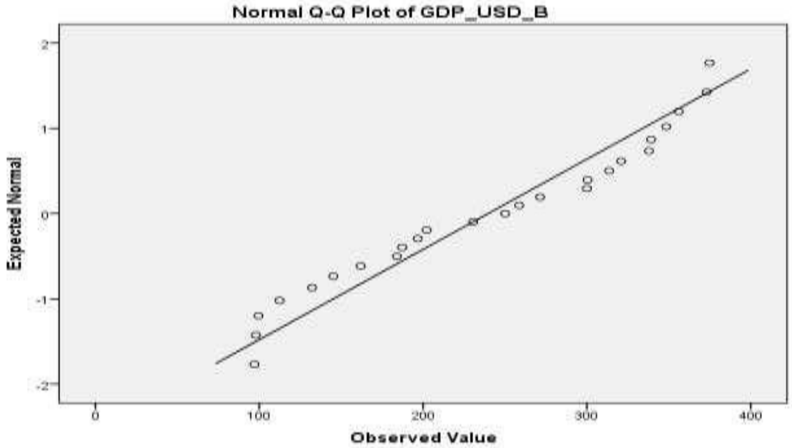
Exchange Rate QQ Plot:



Imports QQ Plot:



GDP QQ Plot:



Explanation:

If the dots closely follow the straight diagonal line, the variable is normally distributed. If they deviate (S-shape, curve, or gaps), it is not normal.

All three Q-Q plots show that the data points lie close to the straight line. This means GDP, Imports, and Exchange Rate all follow a normal distribution, which is good for the regression analysis.

Tests of Normality

	Kolmogorov-Smirnov[a]			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	df	Sig.
GDP_USD_B	.139	25	.200*	.925	25	.068
Imports_USD_B	.119	25	.200*	.950	25	.250
USD_PKR	.154	25	.129	.923	25	.059

*. This is a lower bound of the true significance.

a. Lilliefors Significance Correction

All variables are normally distributed at 5% significance level.

Result: Accept Ho for all variables.

To determine if a data set follows a normal distribution, a normality test was conducted on the transformed data using the **Kolmogorov–Smirnov** and **Shapiro–Wilk** tests in **IBM SPSS Statistics**.

The results indicate that **GDP**, **Imports,** and **Exchange Rate** have significance values greater than 0.05, showing that these variables are normally distributed after transformation.

Techniques:

- Descriptive Statistics
- Correlation Matrix
- Multiple Regression
- Time Series Analysis

Descriptive Statistics:

Descriptive Statistics

	N	Range	Mean	Std. Deviation	Variance	Skewness	
	Statistic	Statistic	Statistic	Statistic	Statistic	Statistic	Std. Error
GDP	25	277.8	239.676	94.3998	8911.317	-.134	.464
Imports	25	72.7	43.496	19.6094	384.530	-.176	.464
USD	25	.4	1.516	.1060	.011	.569	.464
Valid N (listwise)	25						

Ho: Skewness = 0 (data is symmetric/normally distributed)

H1: Skewness not = 0 (data is asymmetric)

	Kurtosis	
	Statistic	Std. Error
GDP	-1.389	.902
Imports	-.525	.902
USD	-.569	.902
Valid N (listwise)		

Ho: Kurtosis = 0 (data has normal tails)

H1: Kurtosis not = 0 (data has light or heavy tails)

Result: Accept Ho for both skewness and kurtosis. This means data is normal enough for analysis.
These two tables show descriptive statistics for three variables over 25 years:

- GDP (GDP in billions of USD)
- Imports (Imports in billions of USD)
- Exchange Rate (Exchange rate: USD to Pakistani Rupee) Key Observations:

- GDP has high variance (8911.3) and negative skew (−1.34) — slightly longer left tail.
- Exchange rate (USD_PKR) has very low variance (0.011) and near-symmetric skew/kurtosis.
- Imports also negatively skewed (−0.176), approximately symmetric.

GDP varies a lot and has a slight left tail, imports are more balanced, and the exchange rate is very stable.

Correlation Matrix:

Correlations

		GDP_USD_B	Imports_USD_B	USD_PKR
GDP_USD_B	Pearson Correlation	1	.965**	.913**
	Sig. (2-tailed)		.000	.000
	N	25	25	25
Imports_USD_B	Pearson Correlation	.965**	1	.876**
	Sig. (2-tailed)	.000		.000
	N	25	25	25
USD_PKR	Pearson Correlation	.913**	.876**	1
	Sig. (2-tailed)	.000	.000	
	N	25	25	25

**. Correlation is significant at the 0.01 level (2-tailed).

Ho: There is no correlation between the two variables (p = 0).

H1: There is a significant correlation between the two variables (p not = 0).

P (rho) = population correlation coefficient

Result: H0 is rejected for all three pairs. There is a statistically significant correlation between each pair of variables at 0.01 level.

- All three variables are strongly, positively correlated with each other (correlations between 0.876 and 0.965).
- All correlations are statistically significant at the 0.01 level (p = .000).
- The strongest correlation is between GDP and Imports (0.965), followed by GDP and USD/PKR (0.913), then Imports and USD/PKR

(0.876).
So basically, over the 25 years studied, Pakistan's economy grew,imports increased, and the rupee weakened against the dollar — all happening together.

Multiple Regression:

Regression Equation:

$$\text{Imports} = \beta^0 + \beta^1 (\text{GDP}) + \beta^2 (\text{USD\_PKR}) + \varepsilon$$

$$\text{Imports} = 3.304 + 0.207 (\text{GDP}) + -6.174 (\text{USD\_PKR}) + \varepsilon$$

Coefficients Table:

Variable	Coefficient	P Value
Intercept (β^0)	3.304 0.207	0.920
GDP (β^1)	-6.174	0.000
USD_PKR (β^2)		0.810

Explanation:

- **β_0 (Intercept) = 3.304** → If GDP and exchange rate are both zero, imports would be 3.30 billion USD. But this is not meaningful in real life, and the p-value is high (0.920), so it's not **statistically significant.**
- **β_1 (GDP) = 0.207, p = 0.000** → When GDP increases by 1 billion USD, imports increase by 0.207 billion USD. This is **statistically significant**.
- **β_2 (Exchange Rate) = -6.174, p = 0.810** → When the rupee weakens by 1 unit (USD/PKR increases), imports decrease by 6.17 billion USD. But the p-value is high (0.810), so this effect is **not statistically significant**. It could be due to chance.

- **R^2 = 0.931** → The model explains 93.1% of the variation in imports. This is very high, meaning GDP and exchange rate together do a great job predicting imports, but mainly GDP is doing the work.

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	USD_PKR, GDP_USD_B[b]	.	Enter

a. Dependent Variable: Imports_USD_B

b. All requested variables entered.

This is the Variables Entered/Removed table from a multiple regression analysis. It shows that both GDP and Exchange Rate were entered together as predictors (independent variables) to explain Imports (dependent variable), using the "Enter" method with no variables removed.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.965[a]	.931	.925	5.3757

a. Predictors: (Constant), USD_PKR, GDP_USD_B

The model summary indicates a multiple correlation coefficient (R) of 0.965, meaning a very strong combined linear relationship between the predictors (GDP, Exchange Rate) and the dependent variable (Imports). The R-squared value of 0.931 shows that 93.1% of the variance in imports is explained by the two predictors. The adjusted R Square

(0.925)* adjusts for the number of predictors, and the standard error of the estimate (5.3757) represents the typical prediction error in billion USD. The model fits the data extremely well.

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	8592.976	2	4296.488	148.678	.000[b]
	Residual	635.753	22	28.898		
	Total	9228.730	24			

a. Dependent Variable: Imports_USD_B

b. Predictors: (Constant), USD_PKR, GDP_USD_B

The model explains no variation in Imports (all population regression coefficients = 0, i.e., $R^2 = 0$)

H1: The model explains significant variation in Imports (at least one coefficient ≠ 0, i.e., $R^2 > 0$)

Conclusion: The overall regression model is significant. Reject Ho

The ANOVA table tests the overall significance of the regression model. The regression sum of squares (8592.976) with 2 degrees of freedom and the residual sum of squares (635.753) with 22 degrees of freedom yield an F-statistic of 148.678, which is significant at $p < 0.001$ (Sig. = .000). This confirms that the model, as a whole, provides a significantly better fit than a null model with no predictors. Overall, the two predictors together significantly predict imports.

Coefficients[a]

Model		Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	Standardized Coefficients Beta	t	Sig.
1	(Constant)	3.304	32.419		.102	.920
	GDP_USD_B	.207	.029	.995	7.250	.000
	USD_PKR	-6.174	25.402	-.033	-.243	.810

a. Dependent Variable: Imports_USD_B

The coefficient (B) = 0 → predictor has no effect on Imports H1: The coefficient (B) ≠ 0 → predictor has an effect on Imports Result: For GDP reject Ho and for USD TO PKR accept Ho. For constant accept Ho.

The coefficients table provides the equation for predicting imports. The unstandardized coefficient for GDP (B = 0.207, p = .000) is statistically significant, indicating that for each additional billion USD of GDP, imports increase by 0.207 billion USD, holding the exchange rate constant. The unstandardized coefficient for USD_PKR (B = -6.174, p = .810) is not statistically significant, suggesting the exchange rate does not independently contribute to the model after controlling for GDP. The standardized coefficient (Beta = 0.995 for GDP) shows GDP dominates the model. The constant term is also non-significant (p = .920). GDP is a strong predictor, and USD_PKR is not significant.

Multiple Regression Summary:

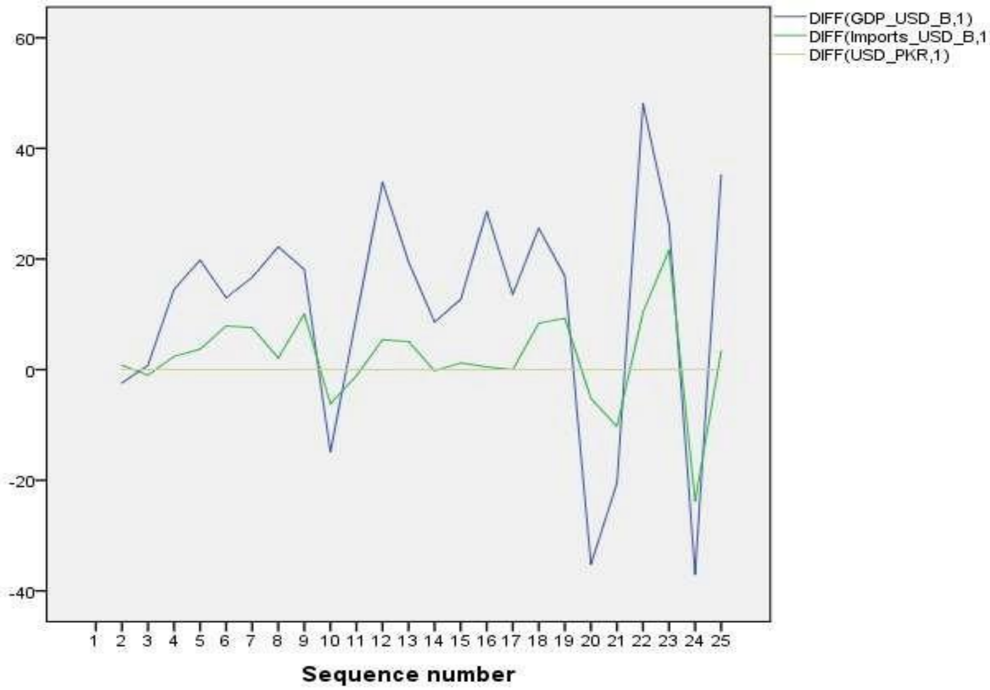
- The model is very good at predicting imports using GDP and the exchange rate.
- But only GDP really matters — the exchange rate doesn't help much.
- The analysis is finished and tells a clear story: GDP drives imports, not the exchange rate.

Interpretation:

- GDP affect imports strongly and significantly.
- The model is good because it explains 93% of import changes (variation).

The exchange rate does not have an independent, significant effect on imports when GDP is considered.

Time Series Analysis:



- **DIFF(GDP, 1)**→ First difference of GDP
- **DIFF(Imports, 1)**→ First difference of Imports
- **DIFF(Exchange Rate, 1)**→ First difference of Exchange Rate

As mentioned earlier, we made the data stationary before forecasting.

- First differencing removes trends and makes data stationary (mean and variance constant over time).
- Stationary data is required for many time series models (ARIMA, VAR, etc.).
- The first-differenced series for GDP, Imports, and USD/PKR fluctuate around zero, indicating that the original data were non-stationary (containing a trend) and differencing successfully removed the trend. The differenced series appear stationary, satisfying the assumption for time series modelling and forecasting.

Autoregressive Integrated Moving Average (ARIMA):

ARIMA is a time series forecasting method. You used it to predict GDP, Imports, and Exchange Rate up to 2030. The models show that GDP and exchange rate have small yearly increases, while imports have no clear yearly trend. The models are clean (no leftover patterns), but they don't predict much better than a simple average because the differenced data is mostly random.

Why it is used?

We had time series data (2000–2024). Normal regression assumes data is independent, but time series data have trends and autocorrelation (today's value depends on yesterday’s). ARIMA handles this.

Model Statistics

Model Statistics

Model	Number of Predictors	Model Fit statistics	Ljung-Box Q(18)			Number of Outliers
		Stationary R-squared	Statistics	DF	Sig.	
GDP_USD_B-Model_1	0	3.331E-16	22.325	18	.218	0
Imports_USD_B-Model_2	0	-2.220E-16	14.550	18	.693	0
USD_PKR-Model_3	0	2.220E-16	20.680	18	.296	0

This table evaluates how well the ARIMA time series model fit the data.

Stationary R-squared ~0.00 means:

- The model does not predict better than a simple mean model on stationary data.
- This is common when we differ in the data.

Ljung-Box Sig. > 0.05 means:

- No significant autocorrelation left in residuals.
- The model captured all predictable patterns.

ARIMA Model Parameters

				Estimate	SE	t	Sig.
GDP-Model_1	GDP	No Transformation	Constant	11.400	4.268	2.671	.014
			Difference	1			
Import-Model_2	Imports	No Transformation	Constant	2.171	1.741	1.247	.225
			Difference	1			
USD/PKR-Model_3	USD/PKR	No Transformation	Constant	.015	.004	3.936	.001
			Difference	1			

This table shows the constant term for each differenced model.

- GDP grows by about $11.4 billion per year on average.
- Imports show no significant trend year to year (p = 0.225).

Exchange rate (USD_PKR) increases by about 0.015 PKR per year on average (slow depreciation).

General ARIMA Model Equation

$$Y'_t = c + \phi_1 Y'_{t-1} + \phi_2 Y'_{t-2} + \cdots + \phi_p Y'_{t-p} + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \cdots + \theta_q \varepsilon_{t-q} + \varepsilon_t$$

Where:

Y'_t	=	Differenced series
c	=	Constant / intercept
ϕ	=	Autoregressive (AR) coefficients
θ	=	Moving Average (MA) coefficients
ε_t	=	Error term at time t

Estimated ARIMA Equations

1. GDP Model — ARIMA(0,1,0)

$$\boxed{\Delta GDP_t = 11.400 + \varepsilon_t} \quad \text{or} \quad \boxed{GDP_t - GDP_{t-1} = 11.400 + \varepsilon_t}$$

Interpretation: The first difference of GDP changes by an average of 11.400 units over time with random error ε_t.

2. Imports Model — ARIMA(0,1,0)

$$\boxed{\Delta Imports_t = 2.171 + \varepsilon_t} \quad \text{or} \quad \boxed{Imports_t - Imports_{t-1} = 2.171 + \varepsilon_t}$$

Interpretation: Imports increase on average by 2.171 units per period with random fluctuations.

3. USD/PKR Exchange Rate Model — ARIMA(0,1,0)

$$\boxed{\Delta(USD/PKR)_t = 0.015 + \varepsilon_t} \quad \text{or} \quad \boxed{(USD/PKR)_t - (USD/PKR)_{t-1} = 0.015 + \varepsilon_t}$$

Interpretation: The USD/PKR exchange rate changes on average by 0.015 units per period along with stochastic error.

Compact Academic Format

$$\boxed{ARIMA(0,1,0):\ \Delta Y_t = c + \varepsilon_t}$$

Thus:

- **GDP:** $\Delta GDP_t = 11.400 + \varepsilon_t$
- **Imports:** $\Delta Imports_t = 2.171 + \varepsilon_t$
- **USD/PKR:** $\Delta(USD/PKR)_t = 0.015 + \varepsilon_t$

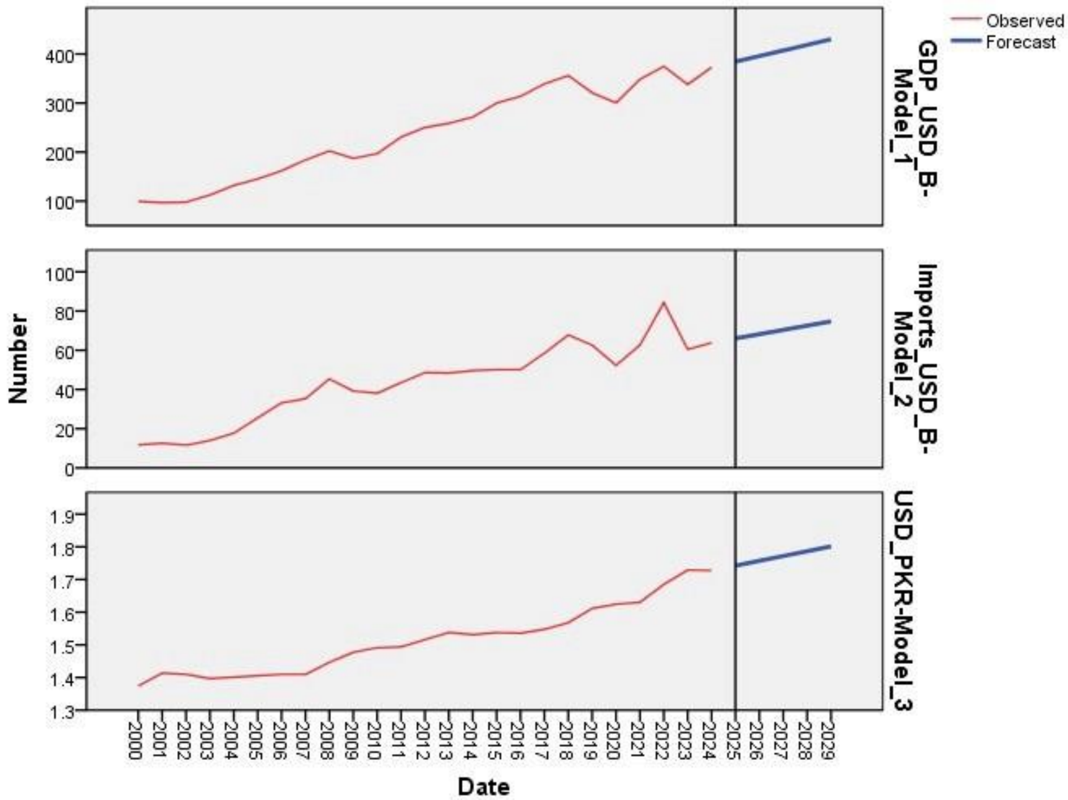
Forecast

Model		2025	2026	2027	2028	2029
GDP_USD_B-Model_1	Forecast	384.5	395.9	407.3	418.7	430.1
	UCL	427.8	457.1	482.2	505.2	526.8
	LCL	341.2	334.7	332.4	332.2	333.4
Imports_USD_B-Model_2	Forecast	66.1	68.2	70.4	72.6	74.8
	UCL	83.7	93.2	101.0	107.9	114.2
	LCL	48.4	43.3	39.8	37.3	35.3
USD_PKR-Model_3	Forecast	1.7	1.8	1.8	1.8	1.8
	UCL	1.8	1.8	1.8	1.9	1.9
	LCL	1.7	1.7	1.7	1.7	1.7

For each model, forecasts start after the last non-missing in the range of the requested estimation period, and end at the last period for which non-missing values of all the predictors are available or at the end date of the requested forecast period, whichever is earlier.

- For each model, forecasts start after the last non-missing in the rangeof the requested estimation period, and end at the last period for which non-missing values of all the predictors are available or at the end date of the requested forecast period, whichever is earlier.
- The time series forecasting analysis was conducted using ARIMAmodels on GDP, Imports, and USD/PKR exchange rate data for Pakistan. Forecasts were generated for the period 2025–2029 after making the series stationary through first differencing. The results indicate a gradual upward trend in GDP and Imports over the forecast period, while the USD/PKR exchange rate remains relatively stable with slight depreciation. Upper and Lower Confidence Limits (UCL & LCL) show the expected forecasting range and uncertainty around future values. Overall, the forecasting models suggest continued economic growth and moderate increases in imports in the coming years.

Forecasting Graphs:



This is a forecast table forecasted forward to 2030 using the time series models.

After making the data stationary, the time series models show that GDP and exchange rate have small but significant yearly increases, while imports do not. The forecasts extend to 2030, but the stationary R^2 values are near zero — meaning the differenced data is mostly random. Therefore, your original (non-stationary) regression results are more useful for real-world interpretation.

Conclusion:

This study found that ***GDP is the main driver of imports in Pakistan***, not the exchange rate. Over 2000–2024, GDP and imports moved strongly together. When GDP increased, imports also increased. The exchange rate (USD to PKR) did not have a significant independent effect on imports after accounting for GDP. The regression model explained 93% of the variation in imports, which is excellent. The time series models confirmed that GDP and exchange rate have small yearly increases, but imports show no clear yearly trend.

Recommendations:

1. Focus on GDP growth policies rather than exchange rate adjustments to manage imports.
2. Do not rely solely on currency depreciation to reduce imports.
3. Promote local production and import substitution industries.
4. Use a mix of trade, industrial, and exchange rate policies together.

Limitations:

1. The study only used three variables, excluding other important factors like inflation and trade policy.
2. The sample size is small (25 annual observations), limiting statistical power.
3. Findings are specific to Pakistan and may not apply to other countries.
   GDP and exchange rate are too closely related (0.913 correlation), which can confuse the model about which one truly affects imports.